

Introduction & Background

Learning and memory are dependent on two key neuromodulatory systems: the dopaminergic (DA) system, which signals the reward prediction error, and the acetylcholinergic (ACh) system, which regulates attention, memory formation, and cognitive encoding. These systems together support how the brain acquires and updates information through error-based learning. Similarly, biologically-inspired machine learning (ML) models have drawn from neuroscience principles to improve efficiency, with algorithms like temporal-difference learning and reinforcement learning that mirror functions of DA and ACh. However, the extent to which these models replicate the specific mechanisms of error-based learning and memory encoding found in the human brain are still unclear — a gap this study aims to address.

Objectives

1. To explore the extent to which biologically-inspired ML models mirror the way DA and ACh systems work in the brain, particularly in relation to learning from errors and forming memories.
2. To investigate whether there are true functional parallels between DA's role in signaling reward prediction errors and the mechanisms used in reinforcement learning algorithms as well ACh's involvement in attention and memory and neural network design.
3. To examine if ML models grounded in such principles demonstrate increased performance.

Methods

Researchers identified relevant articles using the Web of Science Core Collection. The following keywords were implemented to obtain 401 results: machine learning, neurotransmitter systems, and learning and memory. This initial dataset was sorted based on quantity of author citations, with the top 100 most cited articles, reviews, and proceeding papers being taken as eligible studies. From this group, 30 high-impact studies were chosen for a more detailed review to better understand key concepts and findings. These studies were then analyzed to identify similarities between DA and ACh functions in the brain and their corresponding roles in ML models. A comparative approach was used to evaluate how closely biologically inspired ML systems replicate these neural processes, focusing on learning efficiency, adaptability, and memory retention. Bibliometric analysis using Excel and Bibliometrix was completed to identify trends in authorship, institutions, and journals.

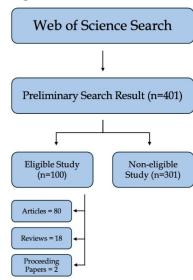


Figure 1: Web of Science Methods

Results

Across the review, several key parallels between biological neuromodulatory systems and ML architectures emerged. Notably, biologically inspired models incorporating replay mechanisms (analogous to hippocampal memory consolidation), neuromodulatory network architectures, and spiking neural dynamics demonstrated improved continual learning, reduced catastrophic forgetting, and greater energy efficiency compared to conventional deep learning approaches. Figure 2 maps the institutional affiliations of the papers, indicating contributions to this field are distributed across a wide range of international institutions rather than concentrated in a few. Furthermore, Figure 3 shows that authorship across the top 100 cited papers is broadly distributed, with Nieder, A. emerging as the only senior author with multiple publications, reflecting the interdisciplinary and dispersed nature of research spanning both neuroscience and machine learning. Whereas Figure 2 encompasses eighteen institutions, Figure 3 lists only one researcher, showing how spread out the research in this field is.

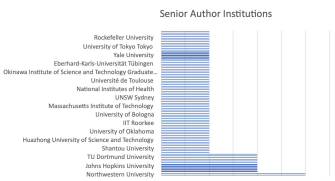


Figure 2: Senior Author Institutional Distribution

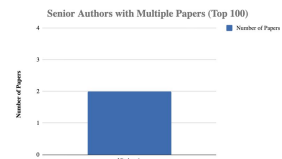
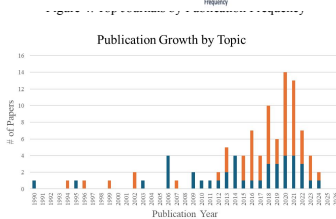
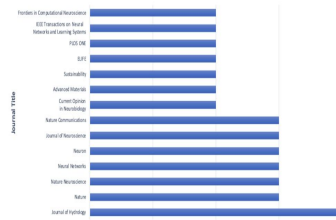


Figure 3: Senior Authors with Multiple Publications



■ # of Neuroscience Papers ■ # of AI/ML Papers

Figure 5: Publication Growth by Topic over Time

Figure 4 identifies journals with multiple top-cited articles, with the Journal of Hydrology (4 papers), Nature, Nature Neuroscience, Neural Networks, Neuron, Journal of Neuroscience, and Nature Communications (3 papers each) leading the field, demonstrating that high-impact work in this space spans both neuroscience-specific and broad-science venues. Figure 5 illustrates publication growth by topic over time, distinguishing neuroscience and AI/ML papers. AI/ML publications show a marked surge beginning around 2013 and peaking in 2020, while neuroscience publications maintain a steadier output across the timeline. The overlap in recent years reflects growing convergence between the two fields and increasing interest in biologically inspired computational models.

Discussion

The findings of our study suggest that biologically inspired machine learning models share strong parallels with the computational roles of dopamine (DA) and acetylcholine (ACh) systems in error-based learning and memory encoding. The similarity between dopaminergic reward prediction error signaling and temporal-difference learning algorithms highlights how neuroscience has already shaped modern ML design, even if that influence is not always explicit. Likewise, the role of ACh in memory encoding parallels attention modulation and synaptic plasticity mechanisms seen in contemporary neural network architectures. Models incorporating replay mechanisms demonstrated improved recall and stability, supporting the idea that biological memory consolidation strategies—such as hippocampal replay during rest—offer practical solutions to persistent limitations in machine learning. However, current models still lack biologically realistic neuromodulatory interactions, particularly the dynamic interplay between DA and ACh systems that likely underpins flexible and adaptive learning in the brain.

Conclusion

Neuromodulatory systems and ML algorithms converge on shared computational strategies for learning and memory. DA encodes reward prediction errors and drives plasticity, while ACh regulates memory encoding and enhances flexibility under changing reward contingencies. These parallels guide the design of AI systems that learn continuously and flexibly, while also informing therapeutic approaches for neuromodulatory disorders. However, current models lack fully realistic NT interactions, and the limited bibliometric scope of 100 papers constrains the generalizability of these findings. In the future, expanding this research to include more recent studies and a wider dataset would improve the reliability of these conclusions. Additionally, using more specific and evolving AI/ML keywords could help capture newer models and applications as the field continues to develop. Further research could also focus on modeling more complex neurotransmitter interactions to better reflect real biological systems.

References